

MA 611: Statistical tools and Computing

Course Code	MA 611
Course Name	Statistical tools and Computing
L-T-P-C	3-1-0-4
Prerequisites	MA-524 (Probability and Statistics) or any course on probability and statistics (like Data Science II) in consultation with the instructor.
Intended for	M.Sc. /M.S./PhD/ B.Tech. 3rd and 4th year
Mutual Exclusion	HS550
Approval	38th Senate (Feb 2023), in the M.Sc Applied Mathematics curriculum; 50th BoA (Jul 2023)

Course Contents

- **Concepts from probability and statistics:** Data (sample vs. Population, histograms, sample mean, median, variance, standard deviation); Probability (axioms, basic rules; and conditional probability); Random variables (discrete vs. Continuous); Review of probability distributions. Some advanced probability distributions: their properties and simulations; Confidence intervals and their significance. (8 Hours)
- **Times series methods:** Collection and classification of data; Different types of diagrams to represent statistical data; Frequency distribution and related graphs and charts; Linear and non-linear models. (6 hour)
- **Regression, classification and multivariate analysis:** Simple regression; Multiple regression; Logistic regression; Generalized linear models; Cross validation; Multicollinearity; Model selection; Prediction and variable selection; Bayesian logistic regression; Principal component analysis; Factors analysis; Discriminant and Classification analysis. (13 Hours)
- **Parametric and Nonparametric tests:** Parametric: Parametric tests are used only where a normal distribution is assumed. The most widely used tests are the t-test (paired or unpaired); ANOVA (one-way non-repeated, repeated), and Pearson rank correlation.
- **Nonparametric:** Non-parametric tests are used when continuous data are not normally distributed or when dealing with discrete variables. Most widely tests used are Chi-squared; Fisher's exact tests; Wilcoxon's matched pairs; Mann-Whitney U-tests; Kruskal-Wallis tests and Spearman rank correlation; Bayesian inference; Kernel Density Estimation. (15 Hours)
- **Laboratory/practical/tutorial Modules:** The labs, using programming languages like R/Python/any other, will take place over a two-hour period in alternate weeks. It will run concurrently with the theory course, thus the subjects for the lab will have previously been established in the theory session.

Text Books

1. James, Witten, Hastie and Tibshirani, **An Introduction to Statistical Learning, with Applications in R**, 2nd edition, Springer, 2021.
2. David D. Hanagal, **Introduction to Applied Statistics: A non-Calculus Based Approach**, Narosa, 2009.

References

1. David Lane, **Introduction to Statistics**, Rice University, David Lane, 2003.
2. Jay Devore, Roxy Peck, Chris Olsen, **Introduction to Statistics and Data Analysis**, 3rd Edition, Wadsworth Publishing, 2008.
3. Peter K. Dunn, Gordon K. Smyth, **Generalized linear models with examples in R**, Springer, 2018.